

Estimating 3D spatiotemporal point of regard: a device evaluation

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This paper presents and evaluates a system and method that record spatiotemporal scene information and location of the center of visual attention, i.e., spatiotemporal point of regard (PoR) in ecological environments. A primary research application of the proposed system and method is for enhancing current 2D visual attention models. Current eye-tracking approaches collapse a scene's depth structures to a 2D image, omitting visual cues that trigger important functions of the human visual system (e.g., accommodation and vergence). We combined head-mounted eye-tracking with a miniature time-of-flight camera to produce a system that could be used to estimate the spatiotemporal location of the PoR—the point of highest visual attention—within 3D scene layouts. Maintaining calibration accuracy is a primary challenge for gaze mapping; hence, we measured accuracy repeatedly by matching the PoR to fixated targets arranged within a range of working distances in depth. Accuracy was estimated as the deviation from estimated PoR relative to known locations of scene targets. We found that estimates of 3D PoR had an overall accuracy of approximately 2° omnidirectional mean average error (OMAE) with variation over a 1 h recording maintained within 3.6° OMAE. This method can be used to determine accommodation and vergence cues of the human visual system continuously within habitual environments, including everyday applications (e.g., use of hand-held devices). © 2022 Optica Publishing Group

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1. INTRODUCTION

Eye-tracking features prominently in studies on object and motion perception [1,2] as well as character and script recognition [3]. In neurophysiology, eye-tracking reveals links between the vestibular and visual systems [4], and transport safety research uses eye-tracking to detect driver fatigue [5]. More recently, there has been growing demand for head-mounted eye-trackers that can be used beyond laboratory studies to investigate behaviors in other real-life scenarios [6,7].

Currently, most commercially available head-mounted eye-trackers use inward-facing video oculography to track the movements of the pupil to map gaze coordinates to images of an outward-facing scene camera (e.g., EyeLink II, TobiiPro Glasses 3, Pupil Labs Invisible, and Core). This mapping of gaze to the scene camera's image allows us to determine areas in the field of view to which users attend with their central (foveal) vision, the area of highest visual awareness. However, conventional video cameras are inherently constrained to two dimensions; from a single image, it is inherently difficult to estimate viewing distance to the point of regard (PoR)—the point in 3D space at which the user is fixating.

Some researchers have made progress in addressing the problem of estimating distance to the PoR using vergence information and stereoscopic displays. For example, Duchowski *et al.* estimated gaze depth from eye vergence along the z axis relative to visual targets simulated at different depths using a Wheatstone stereoscope [8]. Accuracy depended on the dynamic properties of the simulated targets, and optimum performance occurred during stereoscopic presentations of targets that generated some degree of motion parallax. Elmadjin *et al.* proposed an enhanced geometric gaze vector model using a regression-based approach to determine PoR, but concluded that compelling gaze interactions in 3D environments were out of reach due to inaccuracies in the estimation of gaze depth [9]. More recent work estimated vergence eye movements to stereoscopic targets presented on polarizing monitors and suggested that depth cues presented to the foveal periphery influence vergence [10]. However, the reliability of these systems is leveraged on the control of scene information by a rendering software. Hence, it is trivial estimating PoR relative to these targets, as their precise simulated location in depth is known *a priori*.

Estimating PoR relative to objects in the uncontrolled real world poses significant challenges, as the distance to the fixated

target is unknown. One promising way to estimate depth of targets in the scene is by using RGB-D cameras (RGB cameras with a depth sensor, e.g., Microsoft Kinect V2, Intel RealSense Series). Both operating principles of RGB-D cameras (stereo imaging and structured light) require high-contrast images to determine depth in the scene, a constraint that might not be met in real-world environments [11]. Furthermore, the use of projected infrared light patterns as a contrast enhancer of scene images makes these camera systems prone to error, due to contaminant infrared light saturation in the environment that makes them unsuitable for outdoor use [12]. Also, the separation of the eye-tracker and 3D camera (e.g., a hardhat-mounted 3D camera system) generates frequently discarded data and impairs matching of the fields of fixation within the camera's field of view during dynamic real-world activities (e.g., obstructing distance measurements for near-work that is associated with gazing downwards [12–15]). Other research-grade eye-trackers using stereo cameras show high accuracy in matching gaze positions with 3D point clouds. However, these systems have inherent biomechanical constraints due to the size and weight of these typically larger devices as used in studies for extended periods [16,17]. These constraints limit feasibility for the application of these systems in many real-world applications.

To overcome the limitations of previous 3D PoR systems and optimize portability for real-world applications, we developed and validated a new mobile system for spatiotemporal PoR estimation using a modified version of an established eye-tracker. We substituted the scene camera with a lightweight miniature time-of-flight (ToF) camera to map gaze directions to 3D spatial information of the scene. Also, placing a ToF camera in proximity to the estimated cyclopean eye minimizes parallax errors at eye-tracking distances other than the calibration distance [18,19]. The use of a ToF-based eye-tracking system offers great potential for many real-world research applications. For example, the 3D position of the PoR is the main landmark for vergence of the eyes and provides the most valid information about the vergence status of the eyes when considering inter-pupillary distance [20]. Furthermore, the distance to PoR is the main stimulus for accommodation of the eyes, and its cumulative estimation over time can be used to objectively quantify near-work—a metric that has been proposed as a possible risk factor for myopia [21]. Although current technologies can measure near-work objectively, they simply estimate the

directions of gaze and do not measure eye movements [22,23]. Hence, in this study, we propose a lightweight implementation for these applications and evaluate the technology for assessing the 3D PoR in normal human users.

2. METHODS

A. Participants

We recruited 12 participants, including author PW for our study (average age 36.3 years, range 23 to 41 years, eight females and four males) with 10 participants achieving sufficient data quality for the analyses (details in Section 3). In accordance with the Declaration of Helsinki, all participants provided written informed consent, and the study was approved by the Human Research Ethics Committee (HREC) of the University of New South Wales Sydney (HC190522). Vision screening was performed to establish eligibility with respect to the inclusion/exclusion criteria. Adults with good general eye health were included in the study. Exclusion criteria included: monocular visual acuities below 6/7.5 (uncorrected or corrected with contact lenses), presbyopia, binocular vision abnormalities, and unusual pupil findings, including drug use affecting pupil size or shape.

B. Device Description

1. Head-Mounted Eye-Tracker “Core–Binocular” by Pupil Labs

To acquire gaze data, we used the commercially available head-mounted eye-tracker “Core–binocular” (Core Eye-Tracker) from Pupil Labs GmbH, Berlin, Germany [24,25]. The Core Eye-Tracker is equipped with a forward-facing scene camera and two monocular infrared eye cameras [Fig. 1(A)]. To record eye-tracking data, Pupil Labs' open-source software package “Pupil” was used (PLPupil, version 1.21-5). PLPupil uses a glint-free, dark pupil detection method to determine the pupil in each eye-video frame for gaze calculations [25]. In this study, the 3D eye-model mode was used for gaze estimation since it was postulated to compensate for slippage [26]—the undesirable but common displacement of the head-mounted eye-trackers relative to the eye(s). We recorded the eye videos at 120 frames per second and a resolution of 192×192 pixels to optimize

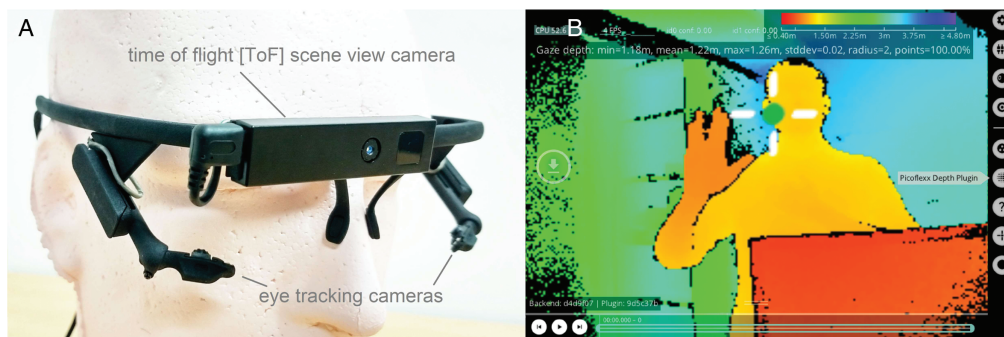


Fig. 1. Configuration of the custom 3D point of regard system and performance. (A) Pupil Labs eye-tracking device “Core–binocular” with Pico Flexx and custom connector. (B) Color coded distance map obtained with PLPupil with Pico Flexx depth plugin showing a sample frame from a social setting with exploratory gaze behavior: red spectrum represents closer distances, and towards the blue spectrum represents farther distances. Black pixels indicate no available distance data. White cross and green circle indicate detected gaze direction.

temporal resolution of pupil tracking. An individual timestamp was recorded along with each video frame, indicating the time of image creation to support temporally nearest time matching of eyes and scene-video frames (see Supplement 1, Section 1). A laptop connected via USB3 powering the Core Eye-Tracker was used as a recording device.

2. CamBoard Pico Flexx Time-of-Flight Camera

To obtain spatial information concerning the arrangement of 3D surfaces in depth within the user's field of view (i.e., physiological field of fixation that subtends all possible gaze directions [27]), we substituted the forward-facing scene camera of the Core Eye-Tracker with the miniature ToF camera "Camboard Pico Flexx" (Pico Flexx), PMD Technologies AG, Siegen, Germany [3D-Eye-Tracker; see Fig. 1(A)] [28]. Pico Flexx records a point cloud, which contains the spatial scene information, with a visual field of 62° horizontal and 45° vertical/ 224×171 pixels (for each pixel, a distance vector is recorded with x , y , and z coordinates)/ $0.28^\circ \times 0.26^\circ$ resolution, respectively. To measure distances over a wide field of view and a range of ambient illumination with high resolution, the Pico Flexx emits modulated infrared light and registers the phase shift of its reflection for each pixel within the sensor's pixel grid to estimate distances. Consequently, radiant power of the Pico Flexx is constant, and only exposure time can be adjusted to achieve an ideal signal-to-noise ratio. Also, the amount of emitted infrared light is limited to specifications for class 1 laser devices. Hence, for the required distance range for this study, to measure distances in dioptric-relevant space of up to 4 to 5 m, the recording frequency is limited to five point clouds per second. Each point cloud is acquired within approximately 30 ms [28]. Thereby, the auto exposure function of the Pico Flexx can regulate exposure time to adjust for objects at closer distances to avoid the overexposure associated with absence of any depth data.

3. Pico Flexx Depth Plugin for PLPupil

To align 3D point cloud data with PLPupil, we developed the custom plugin "PicoFlexx Depth Plugin" (version: Backend: f24533a, Plugin: 0fd0b3f) [29]. One of the major challenges was to acquire raw point cloud data within the constraints of PLPupil's high-frequency event loop. Therefore, the software interface by PMD that accesses point cloud data from the Pico Flexx directly was rewritten to support the temporal efficiency in managing point cloud data [Fig. 1(B)].

C. Study Procedure

1. Calibration of 3D-Eye-Tracker to Match Gaze and Point Cloud Data

Prior to calibrating, the 3D-Eye-Tracker underwent adjustment according to instructions from the manufacturer [30]. The field of view of the scene camera was aligned with the anticipated range of the regions of interest for attentional capture during the study.

Pico Flexx detects a narrow wavelength band of infrared light beyond the emission spectrum of commercial monitors.

Therefore, for calibration, laser printed markers were used since they provide sufficient infrared light contrast to enable the use of PLPupil's "manual marker calibration." For later versions of PLPupil (i.e., 3.0 and above), this is equivalent to "Single Marker Calibration" with marker display mode "Physical Marker" and gaze mapping "3D" [31]. A calibration was considered successful when more than 150 calibration samples were of high quality (e.g., identified marker in the scene camera image and associated gaze coordinate estimated from detected pupils with a confidence value ≥ 0.8) and the PLPupil-reported gaze mapping accuracy was less than or equal to 1.5° . PLPupil calculates the gaze mapping accuracy as the average angular offset between automatically detected fixation targets within the scene camera image and corresponding estimated gaze locations.

2. Study Setup and Execution

From an estimated participant's vantage point, we distributed targets evenly within the Pico Flexx's field of view at distances of 30, 50, 70, 100, and 150 cm (Fig. 2). For inter-participant comparison, the estimated participant's vantage point was aligned with a chin rest, also to reduce possible head movements during testing. All targets were oriented to face the participant's vantage point and subtended the same spatial angle of approximately 5° . All quadrants of the testing field of view contained a closer and a farther distance target. With slight adjustments of the participant's head position, we matched the Pico Flexx's field of view with the study scene. To prevent the overexposure of targets at 30 cm, the Pico Flexx's exposure was set to 810 ns, an empirically determined best fit value for this study.

During accuracy testing, participants were instructed to gaze at the activated green LED at the center of each target. To draw attention to a selected target, the corresponding LED flashed five times and illuminated for 3 s thereafter (total presentation time 4.2 s). To attempt to balance the order-effect of presentation for the intra- and inter-subject distribution of saccade distances, each target was presented prior to, or following, all other possible targets in the set. Resulting repetitions of target presentation added up to a total testing time of approximately 6.5 min.

The sequence of LED activation to guide eye movements was controlled using an Arduino Uno (<https://www.arduino.cc>, Italy). The powering laptop's system time was used to match eye-tracking data and the random LED testing sequence. Inter-target eye movements to transfer gaze were ignored during the first 0.6 s, which is approximately three times longer than the latency of visually guided saccades in healthy normal participants, following the onset of data selection [32]. Hence, up to 18 point clouds per target presentation were classified as containing valid data. As slippage is less likely to be observed within a short period of time [33], we measured the repeatability after approximately 5 min. Furthermore, we repeated the measurements after approximately 1 h to estimate long-term accuracy. At times between repeated measurements, participants were asked to engage in normal everyday office related tasks that may also include near-work with the 3D-Eye-Tracker left on.

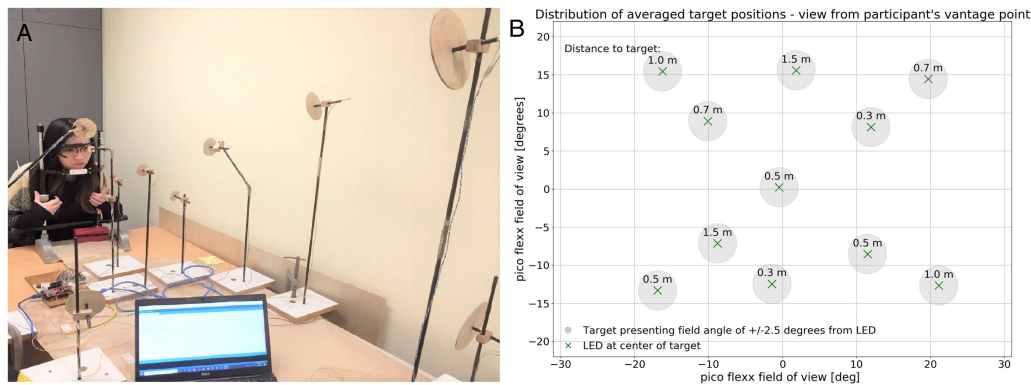


Fig. 2. Study apparatus and target arrangement. (A) Testing setup for gaze accuracy with evenly distributed targets including Arduino controlled green LEDs at center of targets. Targets face predicted observer center in range of chin rest and appear under spatial angle of approximately 5° . (B) Participant views: target distributions averaged from all participants and tests, scaled in degrees of Pico Flexx's field of view.

D. Data Processing and Validation

1. Preprocessing and Statistical Reporting

All data were preprocessed using Python 3.6 (Python Software Foundation, USA) or MATLAB (MathWorks, MA, USA). For statistical data analyses, we used IBM SPSS Statistics 26.0 (IBM Inc., USA). Level of significance was set at 5%.

2. Time Matching of Point Cloud and Gaze Data

PLPupil calculates binocular gaze by alternating gaze estimations between eyes. The effective sampling rate of gaze estimates varies from approximately 120 Hz up to 240 Hz due to fluctuations in temporal linearity of eye image timestamps [33]. Registered gaze timestamps occur approximately 0.11 ± 2.4 ms (sample recording with $n = 2051$, interquartile range of -0.16 to 1.07 ms) after the corresponding scene camera (Supplement 1, Section 1). Considering fixational eye movements with speeds of less than approximately $30^\circ/\text{s}$ only, the error due to lag between gaze and point cloud registration is estimated as very minor.

3. Accuracy of Distance Measurements within Pico Flexx's Field of View

Pasinetti *et al.* analyzed distance measurements in the range of 1.7 to 3.5 m in ideal lighting conditions and achieved accuracies within several millimeters. Performing studies at various distances between 0.3 and 1.5 m in ambient office lighting did not show any variation to their findings [34].

4. Disambiguating Missing Distance Data

Within the Pico Flexx's field of view, approximately 38 k pixels can be registered. Only sufficiently strong signals provide reliable data, depending on the distance to and reflectivity of the object [as presented in Fig. 1(B)]. In general, distance data that are not registered indicate actual object distances of farther than 4.5–5 m. Unfortunately, empirical observations also reveal a small percentage of unavailable distance data due to other

reasons. For example, weak radiance of reflected infrared light due to absorbance properties of object surfaces or close objects shadowing parts of the background due to parallax between Pico Flexx's infrared light emitter and the sensor. Furthermore, overexposure indicating data from very close objects, but no accompanying values for distances, may be generated.

To overcome ambiguity associated with any missing distance data, a filter was designed that used variable radial clustering around detected gaze coordinates to estimate the distance to the PoR. From the point cloud data, within the smallest radius around gaze coordinates (two pixels, 13 pixels total inclusion), the mean distance having a minimum of 20% valid distance data points was assumed to be the ground truth distance to the PoR. Results for which there is no distance data, or data having less than 20% valid pixels within the two smallest radii 2 and 4, are classified as distances beyond 4.5 m [35]. This clustering of pixels was implemented to eliminate possible false distance data of single pixels that might fall onto measured gaze coordinates. Using this approach, the theoretical resolution of any potentially detectable object increases to an approximately 1° to 1.5° field angle. Considering the recording frequency, objects would thus need to be presented for a minimum of 0.2 to 0.25 s within a distance range of 0.2 to 4.5 m for reliable spatial identification.

5. Head Movement Compensation

The participant's head is placed on a chin rest during accuracy testing, and slippage of 3D-Eye-Tracker is assumed to be minor since the overall head movements are very limited, though horizontal and/or vertical head movements in relation to the testing field have been observed. To quantify magnitude of head movements in regard to the testing setup, a custom MATLAB script registered each fixation target position within each recorded point cloud and calculated variations in positions [35]. Head shifts in horizontal and vertical directions have been estimated with magnitudes of up to 4° . The very small inter-target variance of x and y shifts led to the assumption that head rotations around Pico Flexx's z direction were not of significant magnitude.

3. RESULTS

All recordings were assessed for quality using calibration accuracy and reliable gaze data, i.e., gaze data with gaze confidence values above 0.8, a derivative of both pupil detections' confidence. PLPupil reported calibration accuracies for all participants with $\bar{x} = 1.39^\circ \pm 0.28^\circ$ (i.e., ranging between 1.1° and 2.2°). Data from two participants with low performance in calibration ($>1.5^\circ$) also showed a high percentage of unreliable data (gaze confidence <0.8), so these participants were excluded from data analysis. Only a small percentage of unreliable data can be explained with pupil occlusion during blinks but should not exceed 3% to 5% of the overall time. Reviewing eye videos of the two excluded participants indicated that small palpebral apertures led to frequent obstruction of the pupil image, reducing reliability of pupil tracking.

The designed data filter showed a high percentage of point inclusion at gaze coordinates with the smallest radius selected almost exclusively. Overexposure was not observed as intended with the adjustment of exposure. In general, more than 100 measurements per target for each accuracy testing target were determined to have high gaze confidence.

For further analyses, we grouped the targets as a function of eccentricity from Pico Flexx's point cloud center: center target, the single target at center; mid-peripheral, the six targets within angular distance range of 10° to 18° from center; and far-peripheral, the four targets at angular distance of 20° to 25° from center [Fig. 2(B)]. We calculated the magnitude of systematic and random error considering gaze direction (Table 1). The systematic error was calculated as the average gaze deviation from each target center (LED) and random error as the average gaze deviation from the found systematic error for each target [Eqs. (1) and (2)].

Equation (1): systematic error for each target and testing condition:

$$\text{Systematic error} = \left| \frac{1}{n} * \sum_{i=1}^n \begin{pmatrix} x_{i(\text{gaze})} - x_{i(\text{LED})} \\ y_{i(\text{gaze})} - y_{i(\text{LED})} \end{pmatrix} \right|. \quad (1)$$

Equation (2): random error for each target and testing condition:

Random error

$$= \frac{1}{n} * \sum_{i=1}^n \sqrt{(x_{i(\text{gaze})} - x_{(\text{sys error})})^2 + (y_{i(\text{gaze})} - y_{(\text{sys error})})^2}. \quad (2)$$

The systematic error is larger at mid-peripheral and at far-peripheral positions and increased over time by approximately 0.4° for both grouped target positions. Centrally, the random error increased by 0.74° , and in the mid and far periphery by 0.45° for baseline to the 1 h test. Figure 3 shows the systematic and random error distribution for each target with increasing confidence intervals towards the periphery of the Pico Flexx' field of view.

Although systematic and random errors can be computed to estimate the quality of gaze matching, we instead assessed the omnidirectional mean average error (OMAE, i.e., average of the magnitude of angular offset between estimated fixation locations and corresponding locations of the fixation targets) to be a more universally descriptive value to relate gaze estimates to spatial scene information, e.g., minimum resolvable object size the observer is looking at [Eq. (3)].

Equation (3):

$$\text{OMAE} = \frac{1}{n} * \sum_i^n \sqrt{(x_{i(\text{gaze})} - x_{i(\text{LED})})^2 + (y_{i(\text{gaze})} - y_{i(\text{LED})})^2}. \quad (3)$$

To estimate the consistency of measurements, we determined the coefficient of repeatability (95%) (CoR, $1.96 * \text{within participants' standard deviation of the OMAE}$). Including all three visits and all target IDs, the CoR was determined as 2.59° . At center, the CoR was determined with a smaller magnitude of 1.72° , at mid-periphery as 2.57° , and far-periphery with 2.81° . Interestingly, in contrast to the systematic and random error, the CoR of OMAE for the 1 h measurement decreased, compared to the 5 min measurement (Table 2).

We employed a linear mixed model (LMM) to differentiate the main effects of OMAEs for visits, target IDs, and grouped target positions and their interactions. Visits (baseline, repeat after 5 min, and 1 h), target IDs (11 LEDs), and grouped target positions (center, mid-, and far-periphery) were set as fixed factors. Participant ID was used as subject random intercepts to account for non-independent data within each participant. The covariance structure of the random effects was scaled identically. Restricted maximum likelihood was used as the method for

Table 2. Coefficient of Repeatability (95%) of OMAE

Area Time	Center	Mid-Periphery	Far-Periphery	All Targets
Baseline	1.83°	1.76°	2.51°	2.07°
After 5 min	2.00°	3.62°	3.35°	3.40°
After 1 h	1.23°	1.87°	2.48°	2.07°

Table 1. Systematic and Random Errors for Gaze Detection^a

Error Time	Central Error		Mid-Peripheral Averaged Errors		Far-Peripheral Averaged Errors	
	Sys	Ran	Sys	Ran	Sys	Ran
Baseline	1.05°	1.51°	$1.25^\circ \pm 0.71$	$2.02^\circ \pm 0.45$	$1.64^\circ \pm 0.80$	$2.47^\circ \pm 0.67$
After 5 min	1.19°	1.83°	$1.39^\circ \pm 0.77$	$2.32^\circ \pm 0.49$	$1.79^\circ \pm 0.75$	$2.72^\circ \pm 0.38$
After 1 h	1.46°	2.25°	$1.66^\circ \pm 0.91$	$2.47^\circ \pm 0.49$	$1.92^\circ \pm 1.05$	$2.92^\circ \pm 0.53$

^aSystematic error: average deviation angle subtending targeted LED and estimated gaze position [Eq. (1)]. Random error: average deviation angle subtending estimated gaze and average systematic error [Eq. (2)]. Sys, systematic; Ran, random.

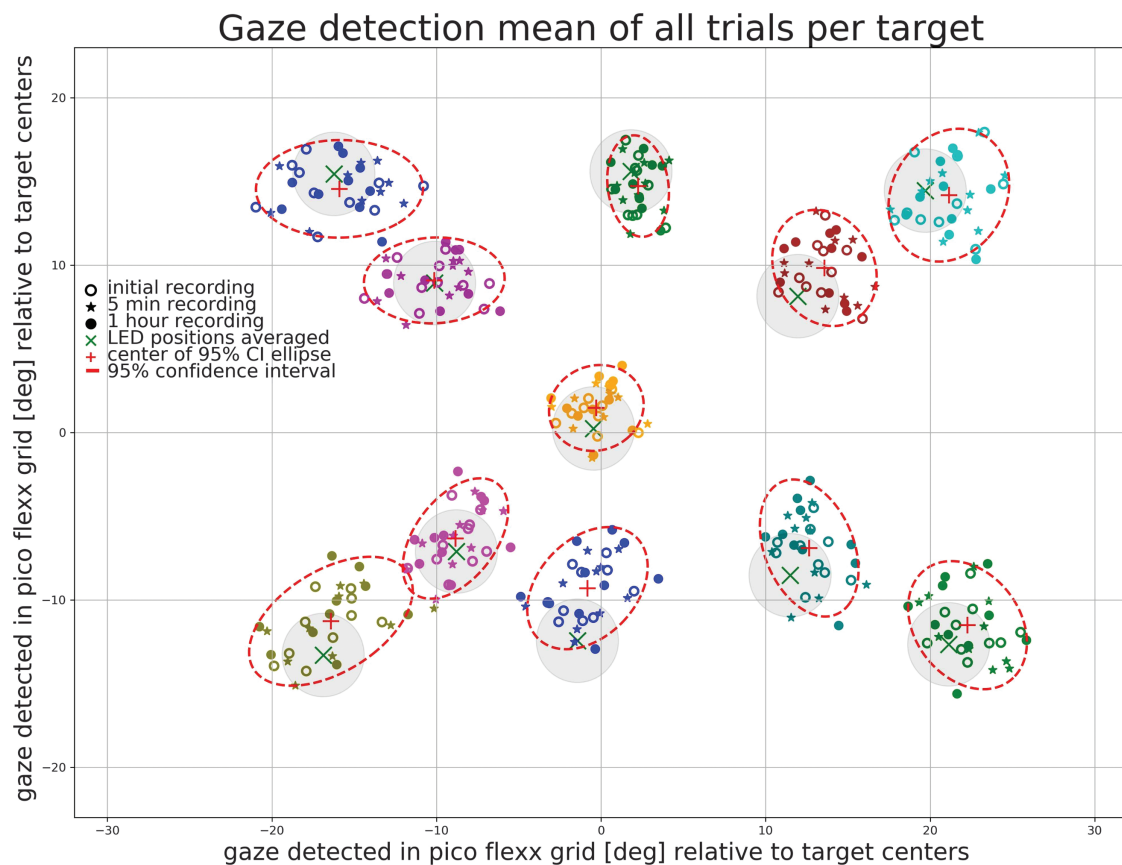


Fig. 3. Mean gaze directions per target. Graph shows measures from all participants at baseline testing: initial recording, short-term repeatability after 5 min, and long-term repeatability testing after 1 h. Means are presented relative to averaged target positions.

estimating the model parameters, and Satterthwaite approximation was used for calculating the degrees of freedom. Multiple comparisons were adjusted using Bonferroni correction.

When comparing OMAEs between visits and grouped targets, the between visits and the interaction of visits and grouped targets did not show a significant difference (LMM, $p = 0.15$ and $p = 0.845$, respectively). In between grouped targets were significantly different, but only for center to far-periphery (LMM, $p = 0.006$). The OMAE model estimates for the grouped targets at center, mid-, and far-periphery were estimated with $\bar{x} = 2.13^\circ$ (CI95% 1.59° to 2.67°), $\bar{x} = 2.72^\circ$ (CI95% 2.41° to 3.04°), and $\bar{x} = 3.00^\circ$ (CI95% 2.66° to 3.35°), respectively.

When comparing OMAEs between visits and individual targets, between visits and between targets showed a significant difference (LMM, $p = 0.03$ and $p < 0.01$, respectively). The interaction between visits and targets was not significant. Also, the significance between visits could not be confirmed with this estimated means model. The means estimation model for individual targets showed differences between various targets. However, the data did not show an obvious pattern of more frequent significance for targets at certain distances or positions within the field of view (Supplement 1, Section 3). The low sample size for each of these conditions limits their practical relevance for further analyses.

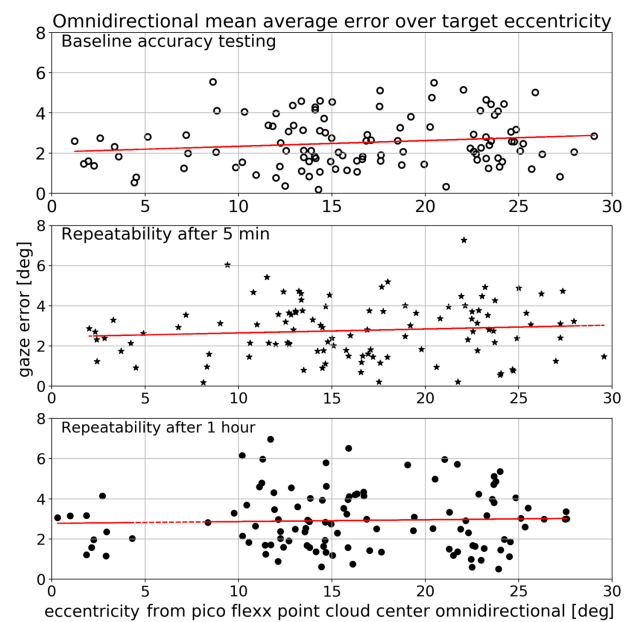


Fig. 4. Error rates as function of fixation eccentricity from Pico Flexx center. Omnidirectional mean average error in regard to eccentricity from Pico Flexx point cloud center of all targets for all participants at baseline (upper), repeatability 5 min (middle), and 1 h testing (lower).

Figure 4 shows the distribution OMAEs for all targets as a function of eccentricity from the PicoFlexx center. With increased eccentricity, the OMAE increases slightly.

4. DISCUSSION

A. 3D-Eye-Tracker Characteristics and Experimental Setup

For this study, we developed a custom Pico Flexx depth plugin that enables head-mounted eye-tracking to monitor scenes in 3D using a TOF camera. The novel device is capable of monitoring visual behaviors in ecological environments. It can record viewing distances over extended periods of time with an acceptable gaze accuracy. The subject's scene view is recorded at a frequency of 5 Hz to achieve the desired dioptrically relevant distance range. Compared to the aforementioned head-mounted eye-trackers with RGB-D scene cameras, the recording frequency of 5 Hz is very slow. However, given the eye's time spent on fixational eye movements (100 ms up to 400 ms) and the restricted disparity of fixations (up to 1.5°), the recording frequency of 5 Hz was sufficient to estimate time spent on viewing test targets at a working range of distances [36]. Even extended saccadic eye movements do not last much longer than 100 ms, but the majority of saccadic eye movements are less than approximately 50 ms [37]. Therefore, it is reasonable to assume that most point clouds are recorded during fixation times. Hence, visual behaviors can be quantified at high resolution. Pico Flexx can record point clouds at higher frequencies with sufficient data quality at closer viewing ranges (if required).

Although the time matching for gaze direction and the scene-view camera is close, gaze detections are more variable and might introduce an error due to the potentially high velocity of eye rotations during saccadic eye movements. In our study, participants fixated on targets for approximately three seconds to sample sufficient data, and a chin rest reduced head movements for inter-participant comparability but did not restrict habitual eye movements. For free-viewing studies, we recommend extracting data from the PoR during fixational eye movements only. The short acquisition time of the point cloud and the corresponding gaze estimation (from temporally nearest gaze data during fixations) should minimize potential errors in 3D PoR estimations for dynamic real-life scenarios. PLPupil offers a fixation detector for *post hoc* analyses.

The placement of more targets within the testing field could potentially identify systematic gaze estimation errors. However, target size and binocular visibility constrain the number of targets for a physically stationary setup.

B. Accuracy of Gaze Mapping onto Point Cloud Data

Core Eye-Tracker can perform at sub- 1° angular error for screen-based applications, within a small field of view of approximately $\pm 10^\circ$, within the calibration area using a 2D mapping method [33]. In real-life scenarios, slippage of any head-mounted eye-tracker is common and introduces errors into gaze direction estimation that need to be compensated for [25]. The ability to detect slippage compromises short-term accuracy as shown in this paper. Slippage detection is based on a mathematical 3D eye model that estimates the eye rotation center location, and the

detection of its displacement is used to compensate for slippage [18,38]. A possible reason for the moderate short-term accuracy might be rooted in the PLpupil 3D eye model that incorporates only a simplified model of the eyes' geometrical properties. The eye's rotation is more complex than a simple concentric rotation of a sphere, and not due only to the movement limiting attachment to the optic nerve [39,40].

Thereby, the eye rotation center has previously been estimated to situate behind the geometric eye center, 0.79 mm nasally from the visual axis and 14.8 mm behind the corneal apex [41]. Other researchers have found the eye rotation center for emmetropes on average was 10.94 mm, with a range of 10.43 to 11.44 mm, behind the corneal pole along the corneal axis, and for myopes, an average of 11.96 mm with a range of 10.58 to 13.37 mm [42,43].

Also, eye cameras do not record the physical pupil but the entrance pupil of the eye, the magnified image of the physical pupil. The entrance pupil plane lies approximately 0.5 mm in front of the physical iris plane, adding to the pupil center's rotational sphere radius. Furthermore, the offset of the visual axis relative to the pupil center is not constant and depends on dilation [44,45]. Changing from mesopic to photopic light conditions (pupil diameter 6.3 to 4.1 mm respectively), the pupil center shifts approximately 0.133 mm nasally with regard to the cornea [46]. This pupil center shift may explain the shift in vergence measurement found by Hooze *et al.* during a study that controlled eye movements and screen illumination changes [47]. Furthermore, pupil center estimations generated from images taken far off the eyes' visual axis generate two additional uncertainties. The estimated pupil center location changes with gaze angle due to its optically induced displacement by corneal refraction and its unknown peripheral asphericity [48,49]. Second, the eye retracts into the orbit during blinks [40,50] and affects vergence measurements immediately prior to and following blink events [51].

The used software version of PLPupil did not support concurrently testing different gaze mapping algorithms nor customizing eye parameters. There are promising algorithms available with later versions of PLPupil [38,52] that could address these limitations if introduced into our system. Nevertheless, the 3D-Eye-Tracker matched gaze onto the 3D point cloud data with the accuracy range proposed by the manufacturer within the calibration area. Further, our study shows that beyond the calibration area the gaze mapping accuracy declines only slightly.

C. Fields of View Matching and Parallax Error

The ergonomic proximity of the Pico Flexx to the eyes appears to offer several advantages. Other 3D eye-tracking systems use a large helmet-mounted RGB-D camera for the scene camera [12,15]. These systems can be used for mid to far distances, but near viewing and inferior fields of fixations are partially obstructed by the camera carrier. The Pico Flexx covers approximately 40% of the total physiological field of fixation, but with fine adjustment of the scene camera's pitch angle prior to calibration, much of the complete range of likely gaze directions can be measured. Indeed, our empirical observations are comparable with previous reports that the majority of gaze directions

are within 20° relative to their overall center and rarely exceed 25° to 30° [16].

The 2D gaze model maps relative shifts in the pupils' centroid in the eye camera images to the scene camera's images during calibration. The proximity of the eyes to the scene camera introduces gaze angle distortion within the scene camera's field of view. Also, the recorded scene camera image is tilted compared to the eye's viewpoint (approximately 1°–1.5° for the angle cyclopean eye's rotation center, center of calibration target, and point cloud vantage point). This image tilt introduces the parallax error with increasing distance to the calibration plane [19]. Instead, the PLPupil 3D gaze estimation model maps gaze vectors onto the scene camera's image by incorporating the distance between the eyes and the scene camera (reducing the parallax error for our system to the point cloud tilt). Therefore, for very close targets, the parallax error becomes more prominent, compared with farther targets of approximately 1.5 m and beyond (see Supplement 1, Section 2). A statistical analysis did not find consistently higher errors of OMAE for near targets when comparing each target individually using the LMM (see Supplement 1, Section 3).

D. Distance Measurement to PoR

Distance measurements throughout the Pico Flexx's field of view show errors within a few millimeters and are considered accurate. Ultimately, these measurements depend on the infrared light reflectivity of the object surface and its angular orientation. The Pico Flexx's auto exposure enables a large range of distances to be measured with a high percentage of valid data (approximately 0.2 to 4 m, equivalent to 5 to 0.25 D optic vergence, respectively). This range is assumed to be the most frequently utilized during near-work [53]. In dioptric terms, the error of a few millimeters corresponds to less than 0.1 D, which is far better than the precision required for refraction measured by eyecare practitioners (0.25 D). Thus, the distance measurement accuracy of this system may be considered very high.

5. CONCLUSION

The results from our present study show that the novel device is capable of measuring distances to the PoR in ecological 3D scenes continuously. The resolution and accuracy of gaze mapping is sufficient to estimate PoR in a range of visual tasks, e.g., when users engage with small hand-held devices. This system will enable vision scientists to measure aspects of gaze in natural viewing conditions, for example, estimating time spent on near-work with greater precision, or inferring the level of a user's spatial central visual attention. This system will offer benefits for assessing 3D visual behavior in a multiverse of real-world industrial applications, including usability analyses and training scenarios.

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Data availability. All code used for analyses with detailed description is available at Github [35]. Open source Pico Flexx depth plugin for PLPupil 1.6 is available at Github [29]. Raw data are available for ethical research on request. Please bear in mind that the entire raw data dataset is 300 GB, and the Pico Flexx point cloud data are accessible only via PMD's Royal software.

Supplemental document. See Supplement 1 for supporting content.

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